2.7.29

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Question

The area of a triangle formed by vertices $\boldsymbol{O}, \boldsymbol{A}$ and $\boldsymbol{B}$, where $\boldsymbol{OA} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\boldsymbol{OB} = -3\hat{i} - 2\hat{j} + \hat{k}$ is

Solution

Let $\boldsymbol{O}$ be the origin
Represent the vectors in matrix form:

$$\boldsymbol{A} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \boldsymbol{B} = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix} \tag{1}$$

The area of triangle OAB is given by:

$$\text{Area} = \frac{1}{2}\boldsymbol{A} \times \boldsymbol{B} \tag{2}$$

Compute the cross product using determinant:

$$\boldsymbol{A} \times \boldsymbol{B} = \begin{pmatrix} (2)(1) - (3)(1) \\ (3)(-3) - (1)(1) \\ (1)(1) - (2)(-3) \end{pmatrix} = \begin{pmatrix} 2 - 3 \\ -9 - 1 \\ 1 + 6 \end{pmatrix} = \begin{pmatrix} -1 \\ -10 \\ 7 \end{pmatrix} \tag{3}$$

Solution

Simplifying:

$$\Rightarrow \boldsymbol{A} \times \boldsymbol{B} = \begin{pmatrix} -1 \\ -10 \\ 7 \end{pmatrix} \tag{4}$$

Magnitude of the cross product:

$$\boldsymbol{A} \times \boldsymbol{B} = \sqrt{(-1)^2 + (-10)^2 + 7^2} = \sqrt{1 + 100 + 49} = \sqrt{150} \tag{5}$$

Final area:

$$\text{Area} = \frac{1}{2}\sqrt{150} \tag{6}$$

C code

```c
#include <math.h>

double OA[3] = {1, 2, 3};
double OB[3] = {-3, -2, 1};

double cross_product[3];

// Function to compute cross product OA x OB
void compute_cross_product() {
    cross_product[0] = OA[1]*OB[2] - OA[2]*OB[1];
    cross_product[1] = OA[2]*OB[0] - OA[0]*OB[2];
    cross_product[2] = OA[0]*OB[1] - OA[1]*OB[0];
}

// Function to compute magnitude of cross product
   vector
double compute_area() {
```

C code

```
compute_cross_product();
    return 0.5 * sqrt(cross_product[0]*cross_product
        [0] +
                         cross_product[1]*cross_product
                            [1] +
                         cross_product[2]*cross_product
                            [2]);
}
```

Python code

```
import numpy as np
import matplotlib.pyplot as plt

# Define vectors OA and OB
OA = np.array([1, 2, 3])
OB = np.array([-3, -2, 1])

# Compute cross product
cross = np.cross(OA, OB)
area = 0.5 * np.linalg.norm(cross)

print(f"Area of triangle OAB = {area:.4f}")

# Plot the triangle vertices and edges
origin = np.array([0, 0, 0])
```

Python code

```
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
# Plot points O, A, B
ax.scatter(*origin, color='black', label='O (0,0,0)')
ax.scatter(*OA, color='blue', label='A (1,2,3)')
ax.scatter(*OB, color='green', label='B (-3,-2,1)')

# Plot edges
ax.plot([origin[0], OA[0]], [origin[1], OA[1]], [
    origin[2], OA[2]], 'b')
ax.plot([origin[0], OB[0]], [origin[1], OB[1]], [
    origin[2], OB[2]], 'g')
ax.plot([OA[0], OB[0]], [OA[1], OB[1]], [OA[2], OB
    [2]], 'r')
```

Python code

```
ax.set_xlabel('X-axis')
ax.set_ylabel('Y-axis')
ax.set_zlabel('Z-axis')
ax.set_title(f'Triangle OAB with Area = {area:.4f}')
ax.legend()
plt.show()
```

Plot

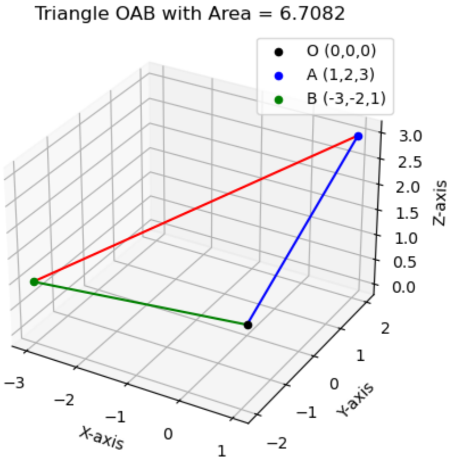

Figure: 3D Plot